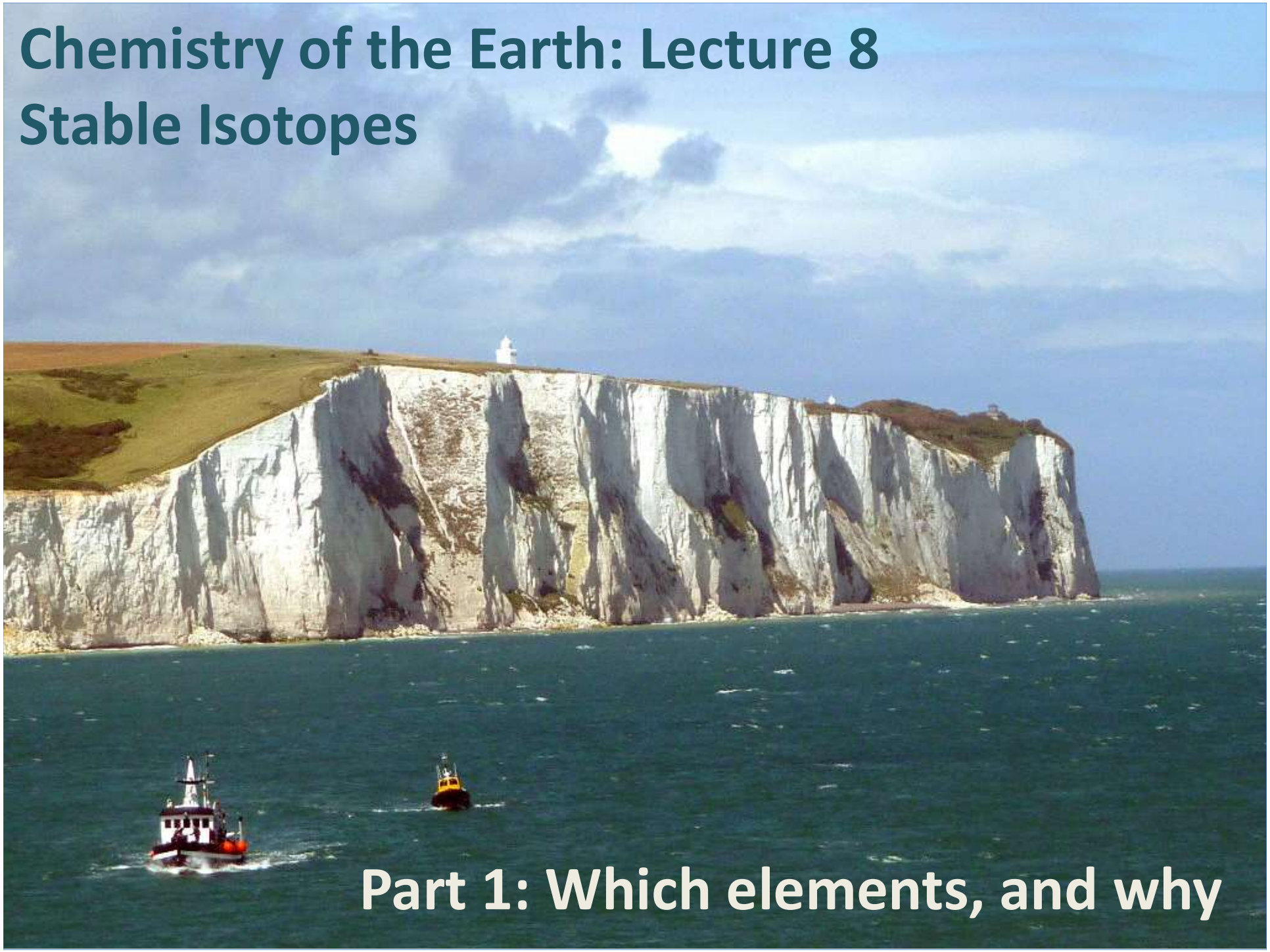
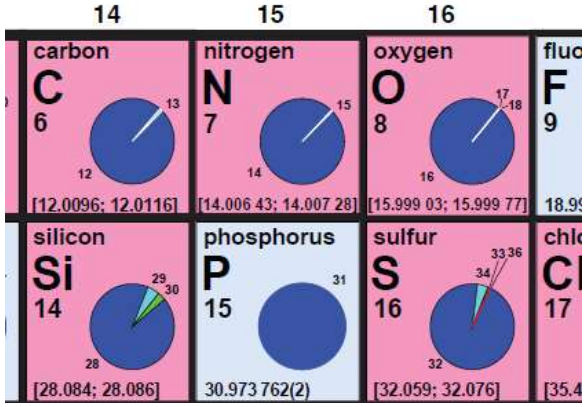
Chemistry of the Earth: Lecture 8
Stable Isotopes

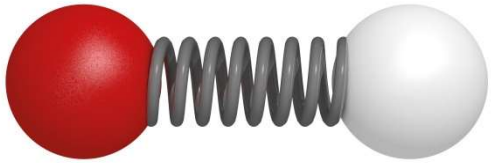


Part 1: Which elements, and why

OBJECTIVES OF THIS WEEK'S LECTURE and PRACTICAL

- Investigate which elements provide the most useful stable isotopes for geochemistry, and why
- Understand how and why isotopes of the same element behave differently in geochemical systems
- Introduction to the terminology for stable isotope systems (α, δ)
- Look at how stable isotopes are used as geothermometers to interpret geochemical processes







READING AND RESOURCES

Grotzinger & Jordan (Basic: Minimum reading)	**p.682-683 (oversimplified)**
Misra (Recommended)	**Chapter 11.3 (p. 253-262)**
McSween & Richardson (Recommended)	**Chapter 13, p. 263-270**
Gill	Chapter 10 (p. 195-197)
White	Chapter 9.1 – 9.4 (p. 371-386)

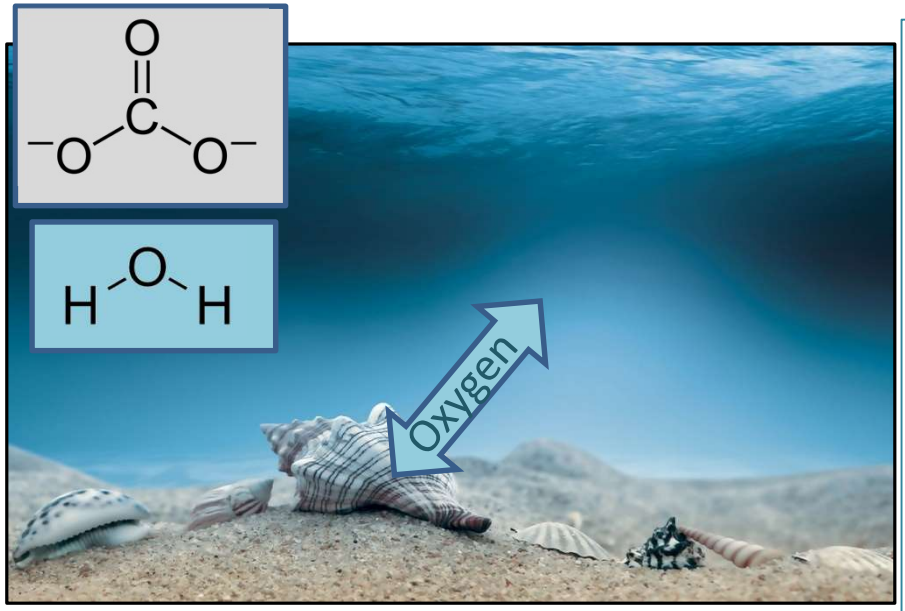
What are stable isotope studies all about?

• In geochemical processes, different isotopes of the same element can behave slightly differently.

• The result is that the different phases in a reaction can have different isotope ratios.

• The separation, or partitioning, of different isotopes is called *fractionation*.

• The basic reason for isotope fractionation is the differences in mass between the isotopes, which in turn affects bond strengths, kinetic energy and other factors.



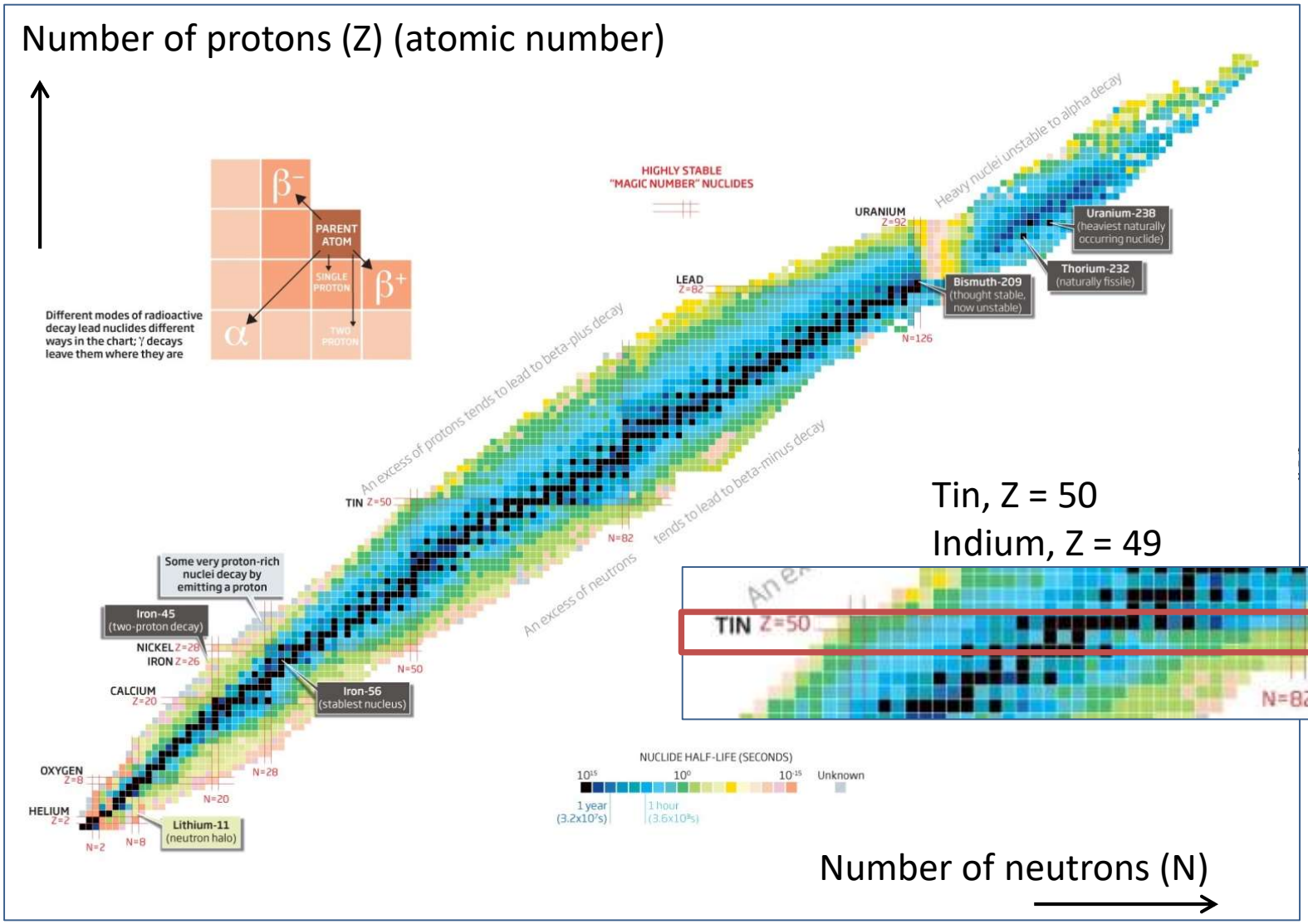
https://wallpaperstock.net/underwater-sea-shells_wallpapers_38260_1920x1080_1.html

Calcite in shells exchanges oxygen with the surrounding water.
This means that isotopes are exchanged:

$$CaC^{16}O_3 + H_2{}^{18}O = CaC^{18}O_3 + H_2{}^{16}O$$

- Oxygen is in different bonding environments
- Calcite favours ^{18}O: $^{18}O/^{16}O$ is higher in calcite
- The ratio $^{18}O/^{16}O$ in calcite changes with temperature

Chart of the nuclides – black squares are stable nuclides



http://www.nuclear-power.net

http://www.nuclear-power.net

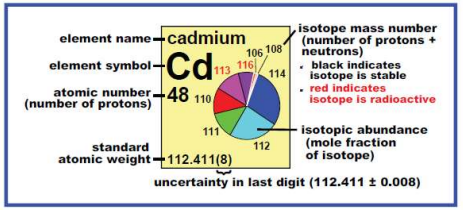
- **For many elements there are multiple stable isotopes**

IUPAC Periodic Table of the Isotopes

Element Background Color Key

[Standard atomic weights are determined using all stable isotopes and selected radioactive isotopes (having relatively long half-lives and characteristic isotopic abundances) in natural terrestrial substances. Isotopes are considered stable (non-radioactive) if evidence for radioactive decay has not been detected experimentally.]

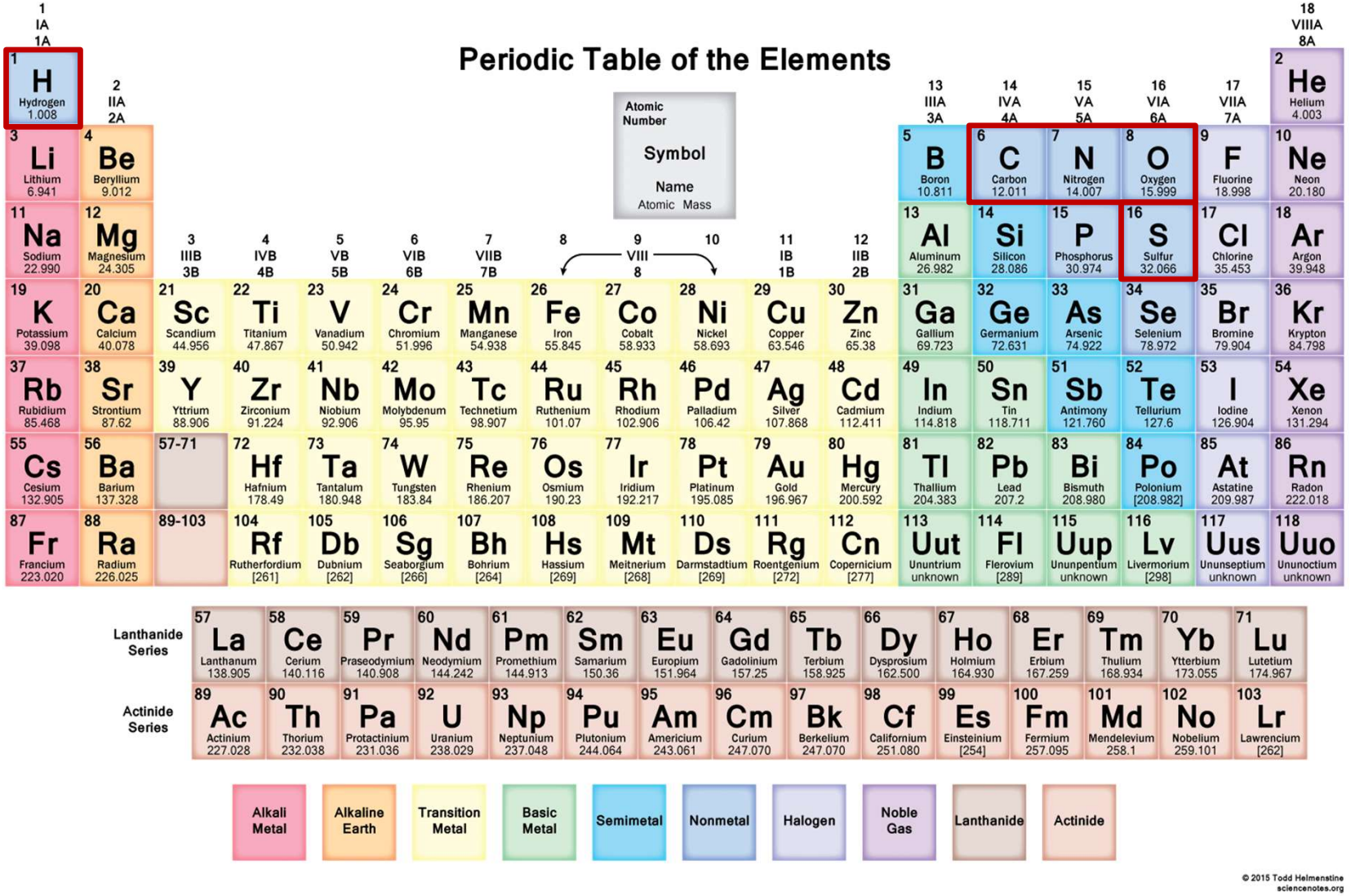
- Element has two or more isotopes that are used to determine its standard atomic weight. The isotopic abundances and atomic weights vary in natural terrestrial substances. These variations are well known, and the standard atomic weight is given as lower and upper bounds within square brackets, [ ].
- Element has two or more isotopes that are used to determine its standard atomic weight. The isotopic abundances and atomic weights vary in natural terrestrial substances, but upper and lower bounds of the standard atomic weight have not been assigned by IUPAC or the variations may be too small to affect the standard atomic weight value. Thus, the standard atomic weight is given as a single value with an uncertainty that includes both measurement uncertainty and uncertainty due to isotopic abundance variations.
- Element has only one isotope that is used to determine its standard atomic weight. Thus, the standard atomic weight is invariant and is given as a single value with an IUPAC evaluated measurement uncertainty.
- Element has no standard atomic weight because all of its isotopes are radioactive and, in natural terrestrial substances, no isotope occurs with a characteristic isotopic abundance from which a standard atomic weight can be determined.

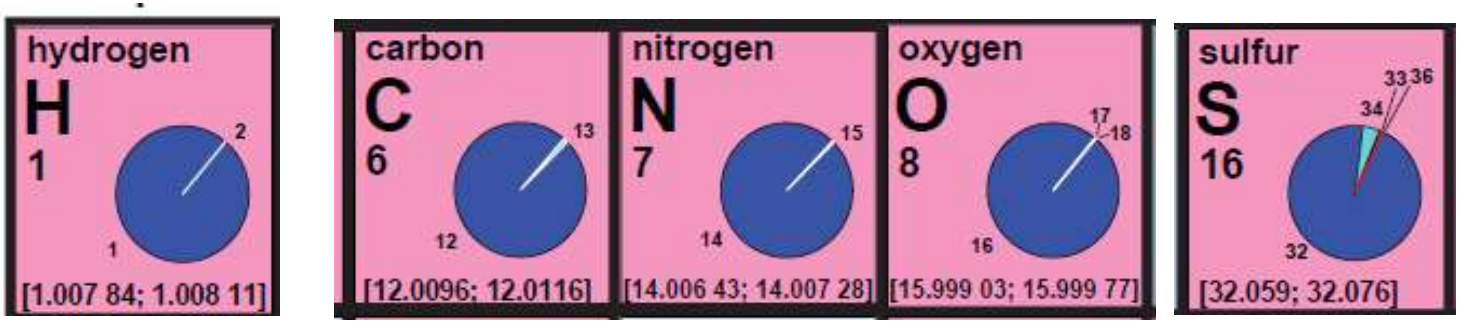


1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
hydrogen H 1 [1.007 84; 1.008 11]																	helium He 2 4.002 602(2)
lithium Li 3 [6.938; 6.997]	beryllium Be 4 9.012 182(3)											boron B 5 [10.806; 10.821]	carbon C 6 [12.0096; 12.0116]	nitrogen N 7 [14.006 43; 14.007 28]	oxygen O 8 [15.999 03; 15.999 77]	fluorine F 9 18.998 4032(5)	neon Ne 10 20.1797(6)
sodium Na 11 22.989 769 28(2)	magnesium Mg 12 24.3050(6)											aluminium (aluminum) Al 13 26.981 5386(8)	silicon Si 14 [28.084; 28.086]	phosphorus P 15 30.973 762(2)	sulfur S 16 [32.059; 32.076]	chlorine Cl 17 [35.446; 35.457]	argon Ar 18 39.948(1)
potassium K 19 39.0983(1)	calcium Ca 20 40.078(4)	scandium Sc 21 44.955 912(6)	titanium Ti 22 47.867(1)	vanadium V 23 50.9415(1)	chromium Cr 24 51.9961(6)	manganese Mn 25 54.938 045(5)	iron Fe 26 55.845(2)	cobalt Co 27 58.933 195(5)	nickel Ni 28 58.6934(4)	copper Cu 29 63.546(3)	zinc Zn 30 65.38(2)	gallium Ga 31 69.723(1)	germanium Ge 32 72.63(1)	arsenic As 33 74.921 60(2)	selenium Se 34 78.96(3)	bromine Br 35 79.904(1)	krypton Kr 36 83.798(2)
rubidium Rb 37 85.4678(3)	strontium Sr 38 87.62(1)	yttrium Y 39 88.905 85(2)	zirconium Zr 40 91.224(2)	niobium Nb 41 92.906 38(2)	molybdenum Mo 42 95.96(2)	technetium Tc 43	ruthenium Ru 44 101.07(2)	rhodium Rh 45 102.905 50(2)	palladium Pd 46 106.42(1)	silver Ag 47 107.8682(2)	cadmium Cd 48 112.411(8)	indium In 49 114.818(3)	tin Sn 50 118.710(7)	antimony Sb 51 121.760(1)	tellurium Te 52 127.60(3)	iodine I 53 126.904 47(3)	xenon Xe 54 131.293(6)
caesium (cesium) Cs 55 132.905 4519(2)	barium Ba 56 137.327(7)	57 - 71 lanthanoids	hafnium Hf 72 178.49(2)	tantalum Ta 73 180.947 88(2)	tungsten W 74 183.84(1)	rhenium Re 75 186.207(1)	osmium Os 76 190.23(3)	iridium Ir 77 192.217(3)	platinum Pt 78 195.084(9)	gold Au 79 196.966 569(4)	mercury Hg 80 200.59(2)	thallium Tl 81 [204.382; 204.385]	lead Pb 82 207.2(1)	bismuth Bi 83 208.980 40(1)	polonium Po 84	astatine At 85	radon Rn 86
francium Fr 87	radium Ra 88	89 - 103 actinoids	rutherfordium Rf 104	dubnium Db 105	seaborgium Sg 106	bohrium Bh 107	hassium Hs 108	meitnerium Mt 109	darmstadtium Ds 110	roentgenium Rg 111	copernicium Cn 112	ununtrium Uut 113	flerovium Fl 114	ununpentium Uup 115	livermorium Lv 116	ununseptium Uus 117	ununoctium Uuo 118

lanthanum La 57 138.905 47(7)	cerium Ce 58 140.116(1)	praseodymium Pr 59 140.907 65(2)	neodymium Nd 60 144.242(3)	promethium Pm 61	samarium Sm 62 150.36(2)	europium Eu 63 151.964(1)	gadolinium Gd 64 157.25(3)	terbium Tb 65 158.925 35(2)	dysprosium Dy 66 162.500(1)	holmium Ho 67 164.930 32(2)	erbium Er 68 167.259(3)	thulium Tm 69 168.934 21(2)	ytterbium Yb 70 173.054(5)	lutetium Lu 71 174.9668(1)
actinium Ac 89	thorium Th 90 232.038 06(2)	protactinium Pa 91 231.035 88(2)	uranium U 92 238.028 91(3)	neptunium Np 93	plutonium Pu 94	americium Am 95	curium Cm 96	berkelium Bk 97	californium Cf 98	einsteinium Es 99	fermium Fm 100	mendelevium Md 101	nobelium No 102	lawrencium Lr 103

Important stable isotopes in geochemistry





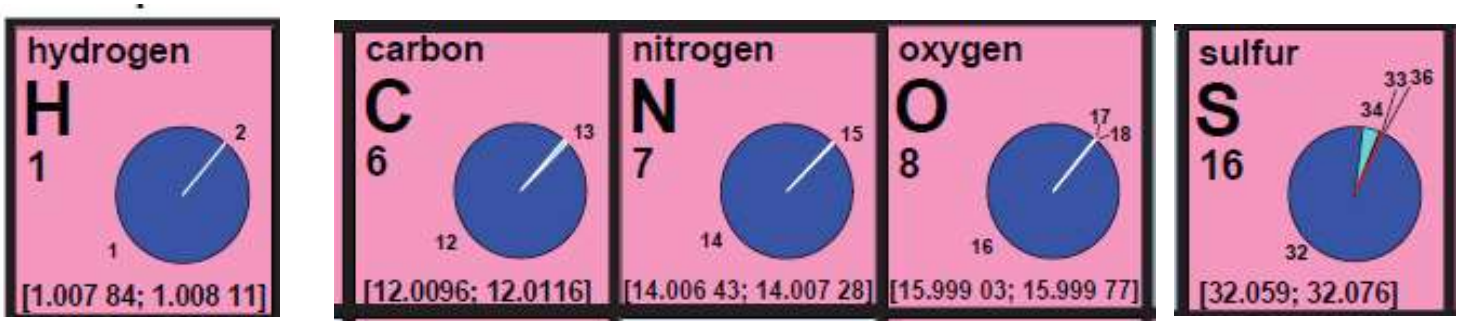
Why are these elements particularly useful for stable isotope studies?

1. The relative difference in mass between the different isotopes of these elements is large. For H, the difference in mass between ^{2}H and ^{1}H is a factor of 2. For C, the difference in mass between ^{13}C and ^{12}C is 8% of the mass of ^{12}C.

2. All these elements are abundant in nature and they are a major portion of common Earth materials. They all occur in common solids, liquids and gases.



Many processes going on here that stable isotopes help to understand



Why are these elements particularly useful for stable isotope studies?

3. C, N and S occur in different oxidation states.

4. All five elements have variable bonding in different materials.

5. The less abundant isotopes occur at measurable levels, making mass spectrometer analyses easier.

6. All five elements are biogenic, i.e. they are important elements in biologic materials. Biologic processes produce big changes in isotope ratios for these elements.



Many processes going on here that stable isotopes help to understand

Isotopes of H, C, N, O, S ("CHONS")

Element	Isotope	Relative abundance (%)
Hydrogen	^{1}H (protium)	99.9844
	^{2}H (D)	0.0156
Carbon	^{12}C	98.89
	^{13}C	1.11
Nitrogen	^{14}N	99.63
	^{15}N	0.37
Oxygen	^{16}O	99.763
	^{17}O	0.0375
	^{18}O	0.1995
Sulfur	^{32}S	95.02
	^{33}S	0.75
	^{34}S	4.21
	^{36}S	0.02

Misra Table 11.1

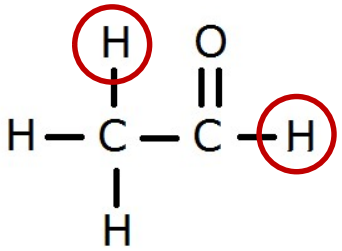
Stable isotope terminology

Isotopologues: Molecules that differ from each other only in isotopic composition e.g. for CO, there are six isotopologues:

$^{12}C^{16}O$, $^{13}C^{16}O$, $^{12}C^{17}O$, $^{13}C^{17}O$, $^{12}C^{18}O$, $^{13}C^{18}O$

Isotopomers: Molecules that differ from one another only in the positions or locations of the isotopic elements:

$CH_3CD{=}O$ $CH_2DCH{=}O$



acetaldhyde

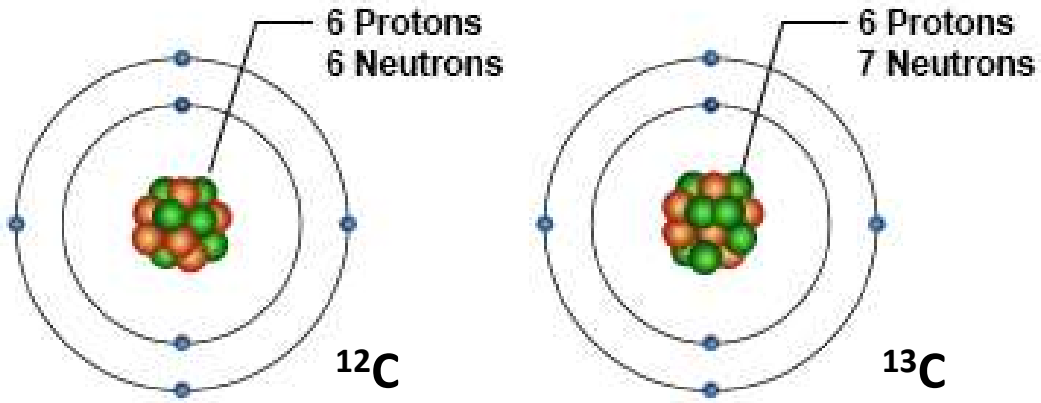
TABLE 13.1. Physical Properties of Water as a Function of Isotopic Composition

	$H_2{}^{16}O$	$D_2{}^{16}O$
Melting point	0.0°C	3.81°C
Boiling point	100.0°C	101.42°C
Maximum density	at 3.98°C	at 11.23°C
Viscosity	8.9 mpoise	10 mpoise

← **Two isotopologues of water**

Why do isotopes of the same elements behave differently?

Isotopes of the same element have the same electron configuration, so we would expect them to have the same chemical behaviour.



However, two different types of behaviour result in isotopic fractionation:

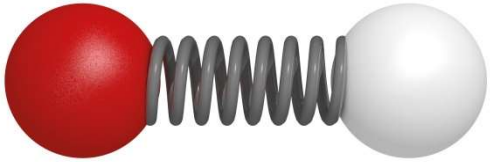
<u>Thermodynamic (equilibrium) mass fractionation</u>

Bonding energies between atoms are different depending on the mass of the atom that is bonding.

<u>Kinetic mass fractionation</u>

Kinetic energy of an atom or molecule is a function of mass (K.E. = ½ m v^2).

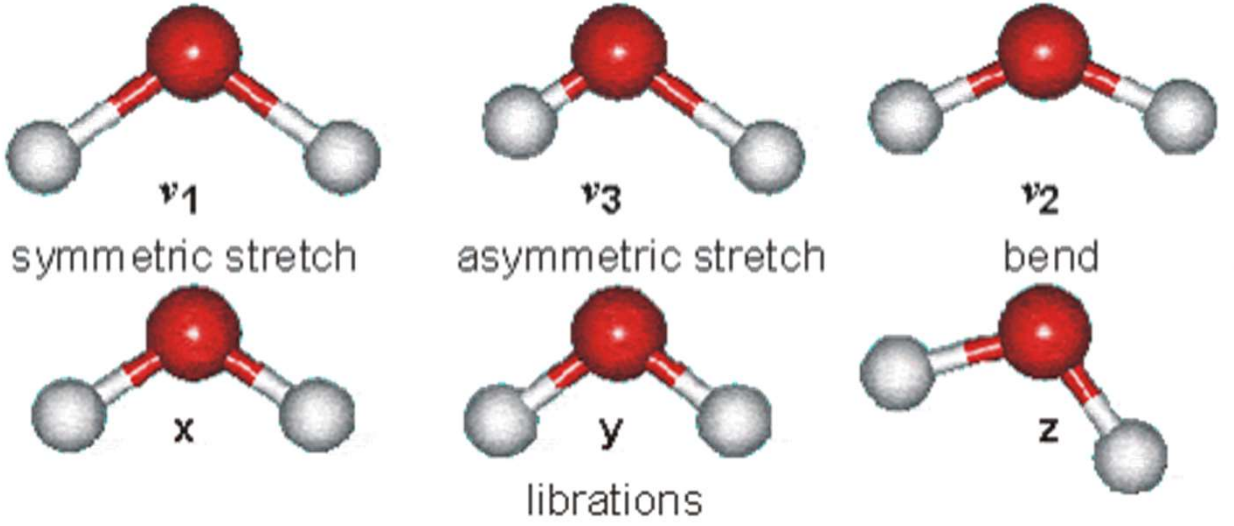
Thermodynamic mass fractionation



• From Lecture 6 on equilibrium:
"Internal energy, U – for a mineral this is the sum of potential energy stored in bonds and kinetic energy of atomic vibrations."

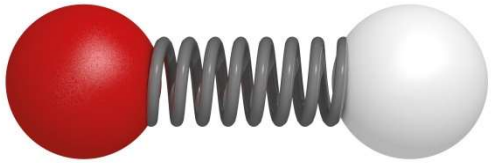
• A bond has a stretching or vibrational frequency. Bonding energies between atoms are different depending on the mass of the atom that is bonding.

Bond vibrations in water



http://u2.lege.net/John_Keely/keelytech.com/wasser.html

Thermodynamic mass fractionation (cont.)



• Vibrational frequency is inversely proportional to square root of mass, so bonds to a lighter isotope have a greater vibrational frequency than bonds to a heavier isotope. Bonds with lighter isotopes have higher internal energy and these bonds are more easily disrupted.

• If a molecule takes up the heavier isotope it will have lower internal energy, and this reduces the energy available for reaction (ΔG), i.e. the phase is more stable.

• **Isotopes fractionate between phases based on the bonding characteristics of each phase.**

Kinetic effects: Where reactions do not reach equilibrium, the reaction products are preferentially light because bonds to light atoms are broken more easily.

Kinetic mass fractionation

Fractionation occurs because kinetic energy is a function of mass.

$$\text{Kinetic energy} = \tfrac{1}{2} m v^2 \qquad (m = \text{mass}, v = \text{velocity})$$

At a given temperature, the kinetic energy of all molecules is the same.
Hence, a molecule with higher mass has a lower velocity.

Differences in velocity lead to:
- differences in diffusion rates
- differences in evaporation rates – faster molecules escape from a surface more easily than slower molecules

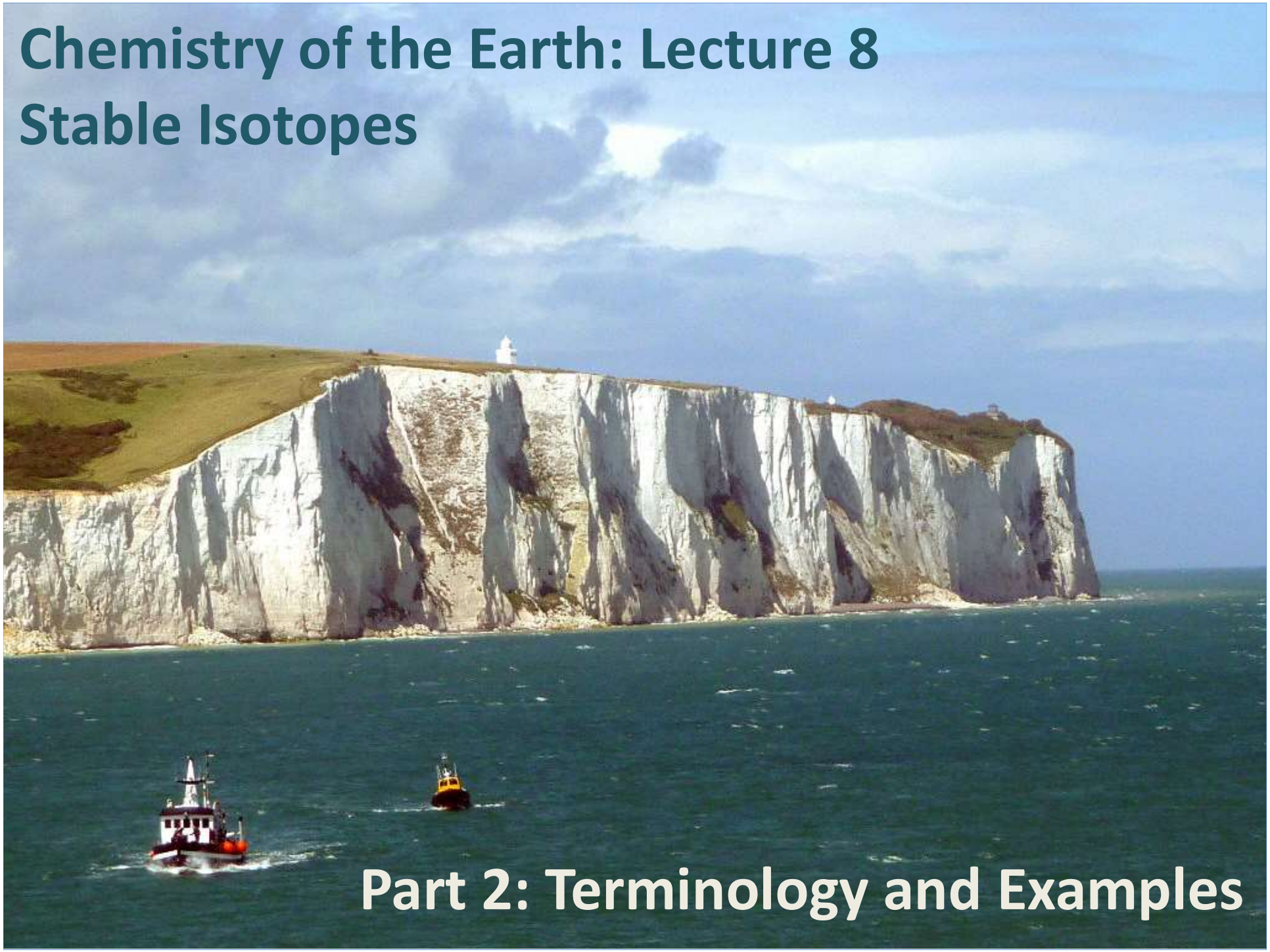
For $^{12}C^{16}O$ and $^{12}C^{18}O$,

$$\tfrac{1}{2} (28) (v_{28})^2 = \tfrac{1}{2} (30) (v_{30})^2$$

$$v_{28} = 1.035\, v_{30}$$

i.e., the average velocity of $^{12}C^{16}O$ molecules is 3.5% greater than that of $^{12}C^{18}O$ molecules. This is true at any temperature.

Chemistry of the Earth: Lecture 8
Stable Isotopes



Part 2: Terminology and Examples

Important terminology for stable isotopes:

Fractionation factor (α)

Describes the distribution of stable isotopes between two phases.

For phases A and B, $\alpha_{A\text{-}B} = R_A / R_B$

- where R is the ratio of the heavy to light isotope.

This is a similar concept to a partition coefficient.

Example: to describe the fractionation of oxygen isotopes ^{16}O and ^{18}O between quartz and magnetite:

$$\alpha_{qz\text{-}mt} = (^{18}O/^{16}O)_{quartz} / (^{18}O/^{16}O)_{magnetite}$$



Quartz

≡



Magnetite

When these minerals are in equilibrium, they have different $^{18}O/^{16}O$ ratios. This is because the bonds to O are different in the two minerals.

Values of α are often close to 1.00X.

Example: the sulfur isotope fractionation ($^{34}S/^{32}S$) between sphalerite (ZnS) and galena (PbS) at 200 °C is 1.00360.

Mathematically, $10^3 \ln(1.00X)$ is approximately similar to X*.

For $\alpha = 1.00360$, $10^3 \ln\alpha = 3.594$

The function $10^3 \ln\alpha$ is used because:

- It's an easier number to speak about, plot, etc. than the values for α.
- There is a smooth relationship between $10^3 \ln\alpha$ and temperature, which is applied to interpret geochemical processes.

*Another way to think of this is that $X = (\alpha-1)*1000$

Plot of $10^3 \ln\alpha$ vs. temperature (°C) - curves

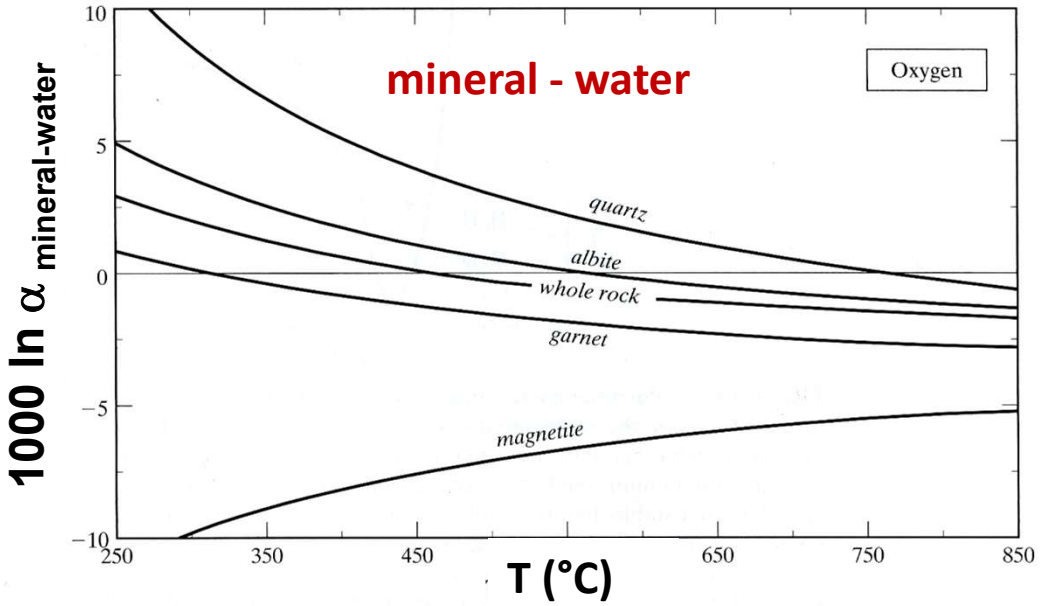
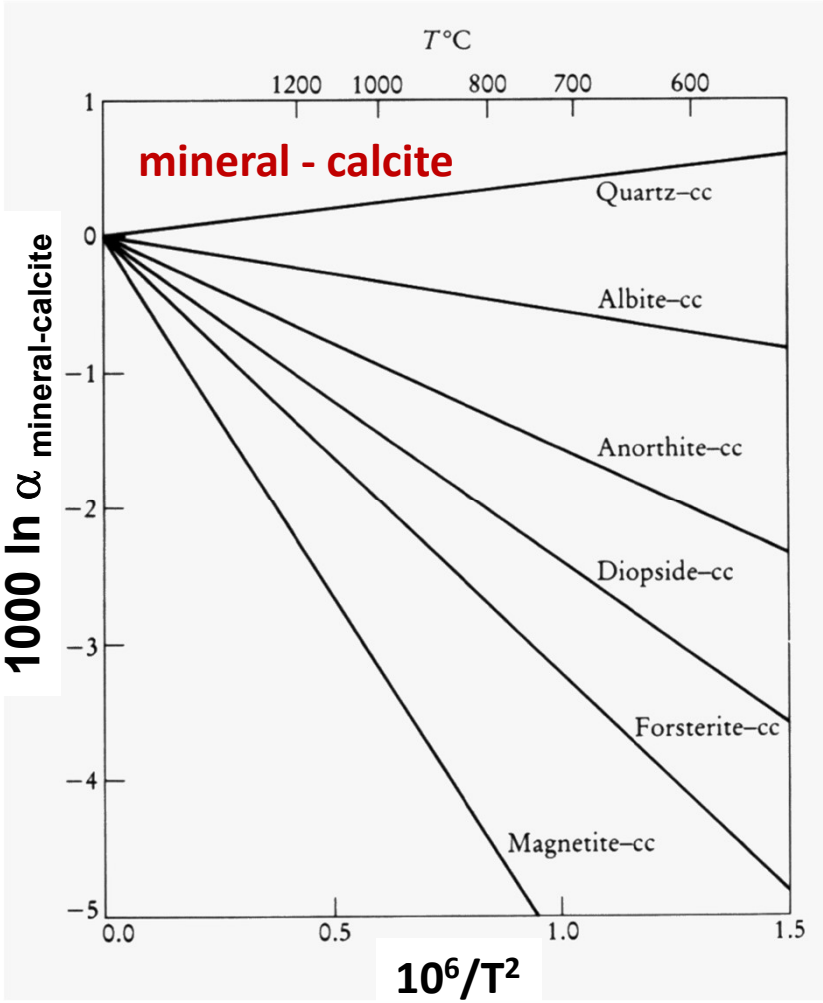


FIGURE 1.4: Oxygen isotope fractionation between selected minerals and water as a function of temperature. Under certain circumstances to be discussed later, knowledge of the $\delta^{18}O$ values of two cogenetic phases allows the temperatures of formation to be determined.
Note: $1{,}000 \ln\alpha_{a-b} \cong \delta^{18}O_a - \delta^{18}O_b$ for phases *a* and *b*.

Sharp

Plot of $10^3 \ln\alpha$ vs. $10^6/T^2$ (T in K) - linear



Rollinson

Important terminology for stable isotopes:

Delta notation (δ)

It is difficult to measure absolute values of isotope ratios, so measured values are compared to a standard.
The delta notation is a way of comparing an isotope ratio to the ratio measured in a standard.

$$\delta = \frac{(R_{sample} - R_{standard})}{R_{standard}} * 1000 \ (‰)$$

R is the ratio of the heavy to light isotope

Or..... $$\delta = \left[\frac{R_{sample}}{R_{standard}} - 1 \right] * 1000 \ (‰)$$

Per mil ‰
Parts per thousand

Per cent %
Parts per hundred

Note that if we measure the standard as the sample, $\delta = 0$

Delta notation (δ)

$$\delta = \left[\frac{R_{sample}}{R_{standard}} - 1 \right] * 1000 \ (‰)$$

Per mil ‰
Parts per thousand

Per cent %
Parts per hundred

Typical differences in isotope ratios (R values) in natural processes are at a level of parts per thousand.

For example, let's consider the ratio $^{18}O/^{16}O$, and let's say a sample has an amount of ^{18}O one part per thousand higher than the natural abundance:

Natural abundances

$(^{18}O/^{16}O)_{sample} / (^{18}O/^{16}O)_{standard}$ = (**0.1996** / 99.763) / (0.1995 / 99.763)

= 0.002001 / 0.002000

= 1.0005

Subtract 1 → 0.0005 = 0.5 ‰ = 0.05 %

In other words, the sample has $^{18}O/^{16}O$ that is 0.5 ‰ or 0.05 % higher than the $^{18}O/^{16}O$ ratio in the standard.

$$\delta = (R_{sample} - R_{standard}) / R_{standard} * 1000 \quad ‰$$

For oxygen,

$$\delta^{18}O = (({}^{18}O/{}^{16}O)_{sample} - ({}^{18}O/{}^{16}O)_{SMOW}) / ({}^{18}O/{}^{16}O)_{SMOW} * 1000 \quad ‰$$

SMOW = Standard Mean Ocean Water, a universal standard for O and H.

δ can be positive or negative: A positive value means the sample has a higher heavy/light isotope ratio than the standard, and a negative value means the sample has a lower heavy/light isotope ratio than the standard.

Using the same example as before:

$({}^{18}O/{}^{16}O)_{sample} / ({}^{18}O/{}^{16}O)_{standard}$ = (**0.1994** / 99.763) / (0.1995 / 99.763)
= 0.001999 / 0.002000
= 0.9995
Subtract 1 → -0.0005 = -0.5 ‰ = -0.05 %

In other words, the sample has ${}^{18}O/{}^{16}O$ that is 0.5 ‰ or 0.05 % **lower** than the ${}^{18}O/{}^{16}O$ ratio in the standard.

Putting δ together with α:

$10^3 \ln\alpha$ is approximately the same as $\delta_A - \delta_B$

$$10^3 \ln\alpha \approx \delta_A - \delta_B = \Delta_{A\text{-}B}$$

Example - calculation of α

At 10 °C, $\delta^{18}O$ for liquid water, $\delta^{18}O_{lw}$ = -0.80 ‰

$\delta^{18}O$ for water vapour, $\delta^{18}O_{wv}$ = -10.79 ‰

$$\Delta_{lw\text{-}wv} = \delta_{lw} - \delta_{wv} = -0.80 - (-10.79) = +9.99 \text{ ‰}$$

$$1000 \ln \alpha = 9.99$$

$$\alpha = \exp(9.99 / 1000) = 1.01004$$

Another example – calculation of α

Consider the equilibrium between CO_2 in the atmosphere and bicarbonate ions, $[HCO_3^-]$, in a sample of river water.

Atmospheric CO_2 has a $\delta^{13}C$ value of -7 ‰.

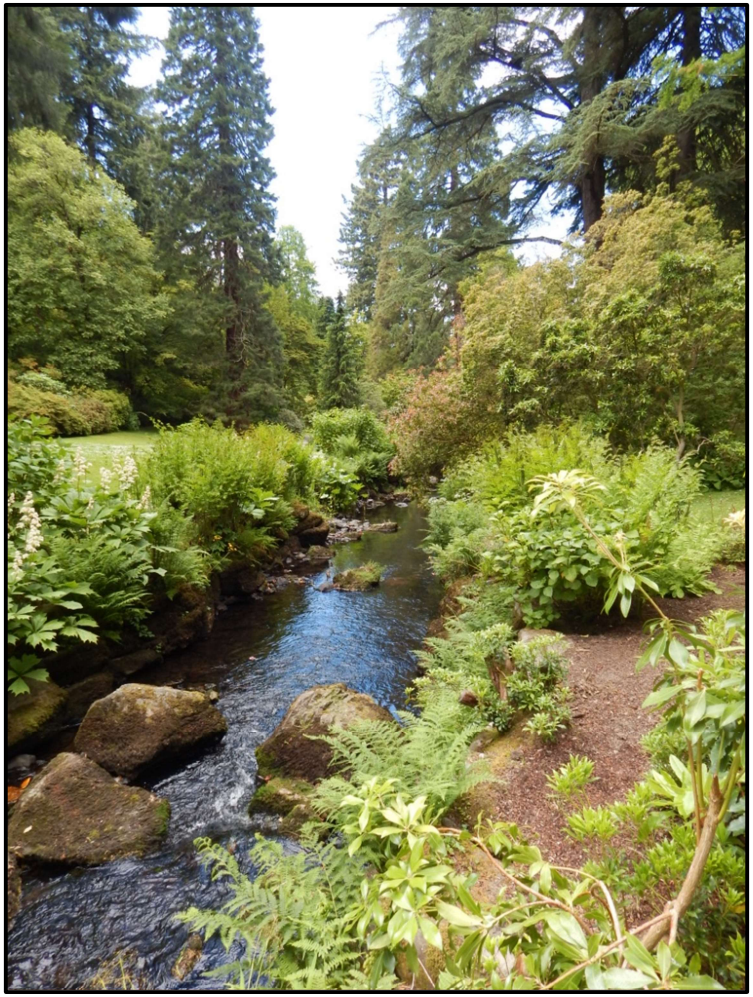
$[HCO_3^-]$ ions have a $\delta^{13}C$ value of +1.24 ‰.

What is the fractionation factor, $\alpha_{(HCO_3^-)-(CO_2)}$?

$$1000 \ln \alpha_{(HCO_3^-)-(CO_2)} = \delta_{(HCO_3^-)} - \delta_{(CO_2)}$$

$$1000 \ln \alpha = 1.24 - (-7) = \mathbf{8.24}\ ‰$$

$$\alpha = \exp(8.24 / 1000) = 1.00\mathbf{827}$$



Stable isotopes: Geothermometry

Geothermometry – determining the temperature of a mineral reaction. Very useful for a wide range of geologic applications.

Example: (Original idea – H. Urey). If we want to examine the geologic record of climate change, we would like to know the temperature of ancient oceans.



Benthic foraminifera

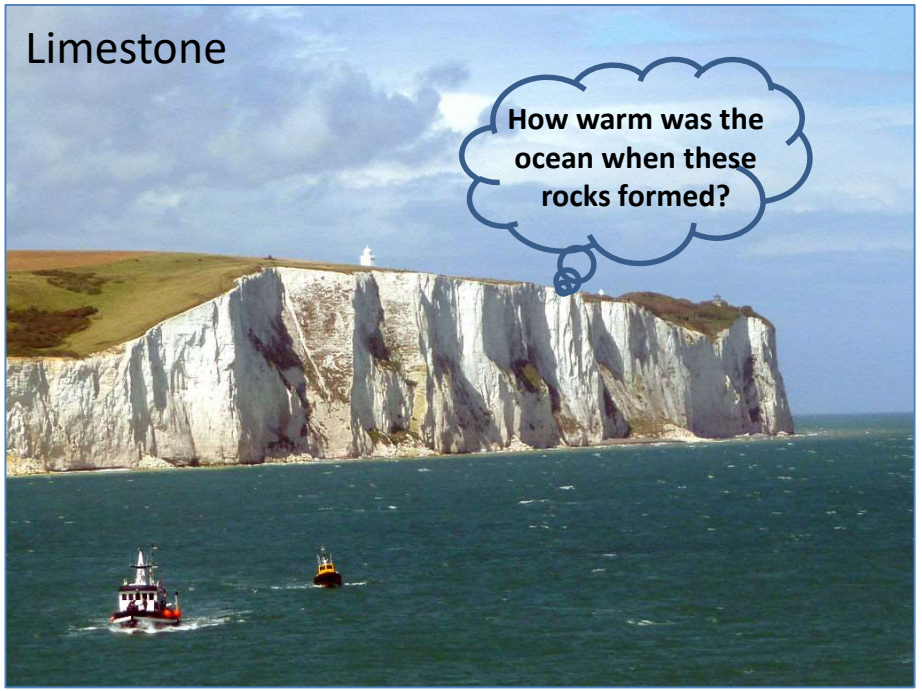
One way to approach this is to look at isotopic exchange in the equilibrium between calcite ($CaCO_3$) and water (H_2O):

$$CaC^{16}O_3 + H_2{}^{18}O \;=\; CaC^{18}O_3 + H_2{}^{16}O$$

This equilibrium reaction shows only isotopic exchange, not chemical change.

Calcite: in organisms such as benthic foraminifera

Water: ocean water
(SMOW = standard mean ocean water)



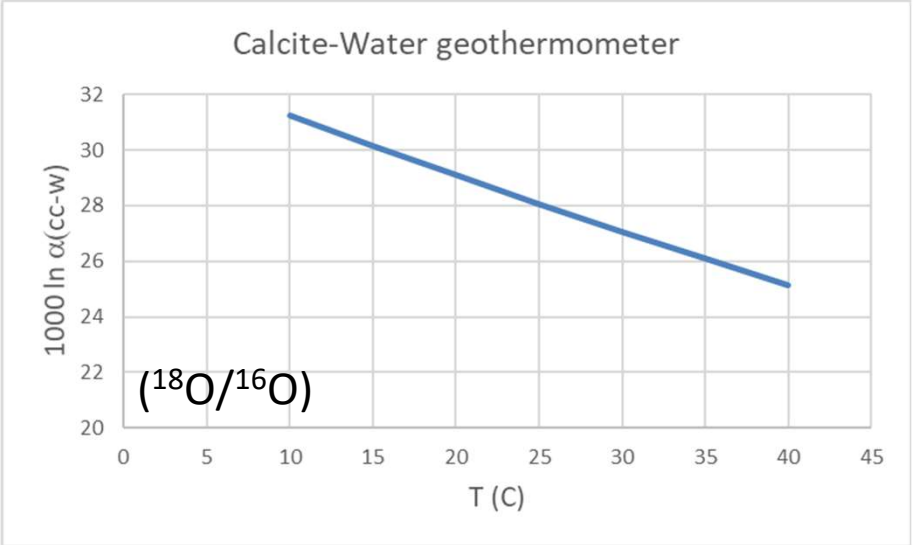
https://commons.wikimedia.org/wiki/User:Immanuel_Giel

For oxygen isotope exchange between calcite and water, $\alpha_{(calcite\text{-}water)}$ is a function of temperature:

$1000\ln\alpha_{(Calcite\text{-}H20)} = 18.03(10^3\,T^{-1}) - 32.42$

Example:
20 °C, $1000\ln\alpha_{(Calcite\text{-}H20)} = 29.08$

25 °C, $1000\ln\alpha_{(Calcite\text{-}H20)} = 28.05$



In principle, if we measure $\delta^{18}O$ in limestone, we can calculate the temperature of the seawater when the calcite formed.

This simple concept is difficult to achieve in practice, because the value of SMOW changes e.g. during glacial / interglacial periods (more of that next week). But it illustrates how stable isotopes can be used for palaeoclimate studies.



https://commons.wikimedia.org/wiki/User:Immanuel_Giel

Stable isotopes: Geothermometry using mineral pairs

At high temperatures, $1000 \ln \alpha_{A\text{-}B} \approx A / T^2$ (T in Kelvin, A is a constant)

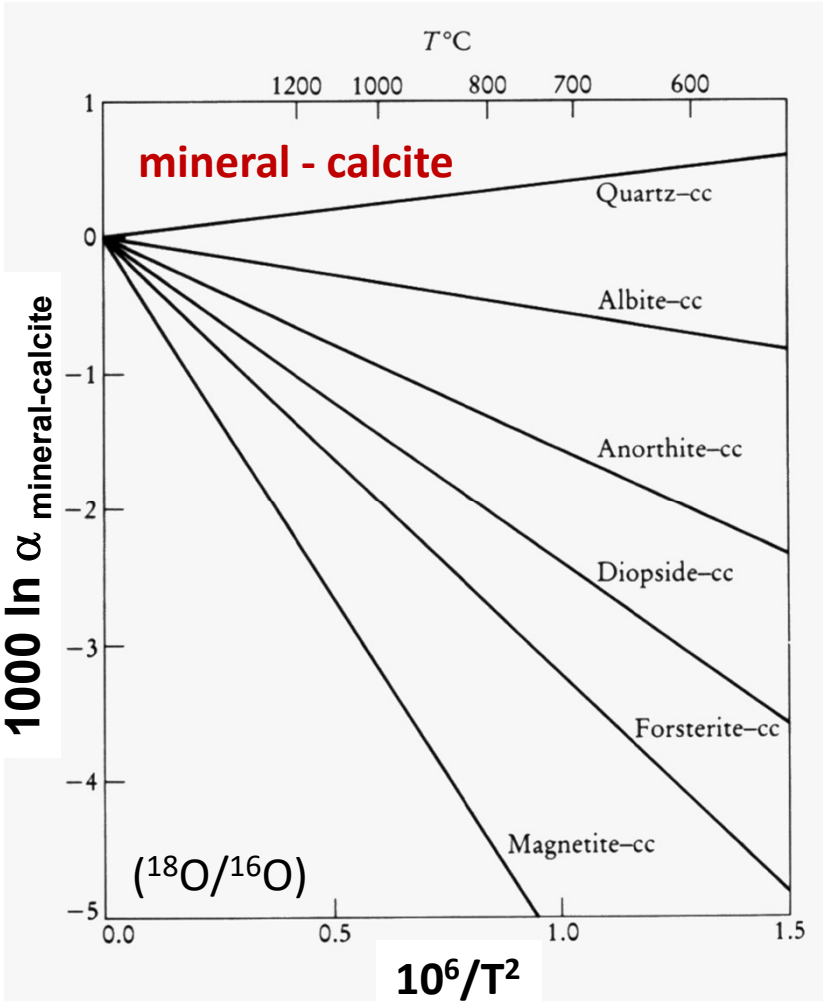
i.e. a straight line on a plot of $1000 \ln \alpha_{A\text{-}B}$ vs. $1/T^2$

The temperature dependence of α is the basis for stable isotope geothermometry.

At high temperatures, α approaches a value of 1, i.e. no fractionation (ln1 = 0).

So this technique works best for low-temperature equilibria where fractionations are larger.

Stable isotope geothermometry mostly uses isotopes of O and S.



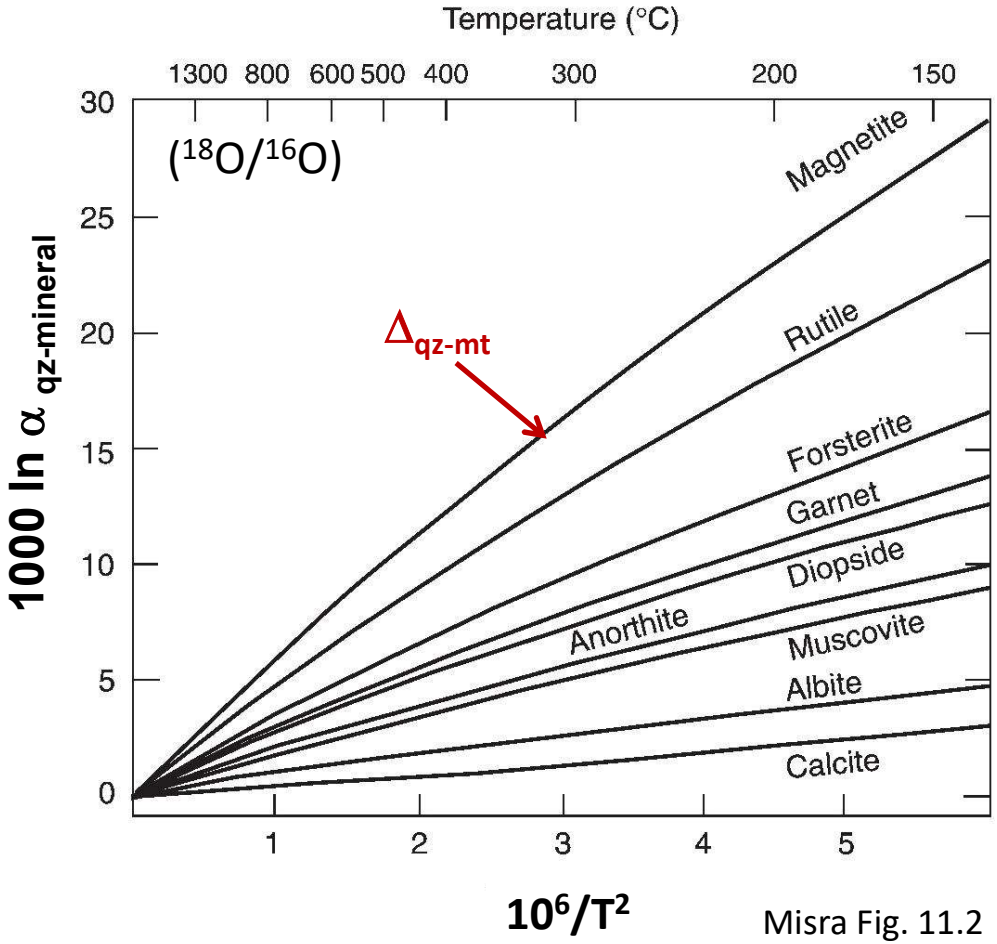
Oxygen isotope geothermometry – silicate and oxide minerals

Figure shows oxygen isotope fractionation factors between quartz and various other minerals as a function of $1/T^2$.

Example: For the mineral pair quartz – magnetite (qz-mt),

$$\Delta_{qz\text{-}mt} = 1000 \ln \alpha_{qz\text{-}mt}$$

At equilibrium, ^{18}O is preferentially partitioned into quartz relative to all other minerals, so all the values of 1000 lnα in the plot are positive.



Misra Fig. 11.2

- Quartz-magnetite is a useful geothermometer – big values of α.
- Quartz-feldspar is not a useful geothermometer – small values of α.

Sulphur isotope geothermometry – sulphide minerals

CDT = Canyon Diablo Troilite

For sulphur, we consider ^{34}S and ^{32}S :

$$\delta^{34}S = (((^{34}S/^{32}S)_{sample} - (^{34}S/^{32}S)_{CDT}) / (^{34}S/^{32}S)_{CDT}) * 1000 \text{ ‰}$$

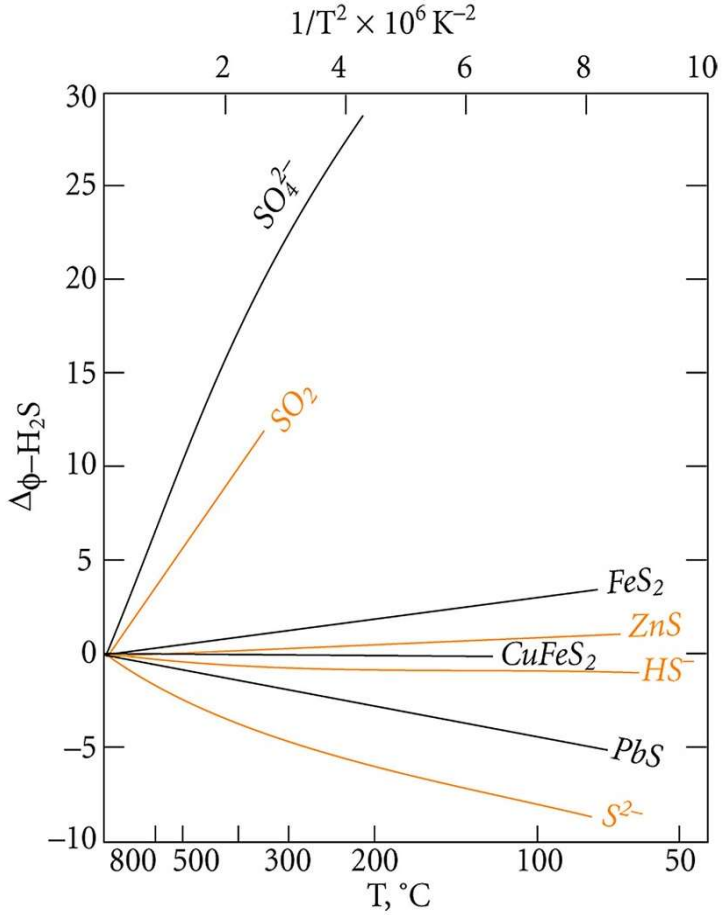
$$\Delta^{34}S = \delta^{34}S_{min1} - \delta^{34}S_{min2}$$

The most commonly used S isotope geothermometer is sphalerite – galena. Used to determine the temperature of ore formation.



Sphalerite (ZnS) with galena (PbS) and chalcopyrite ($CuFeS_2$). Huaron Mine, Peru.

geology.com/minerals/sphalerite.shtml



Fractionation of S isotopes between S species and H_2S.

Chemistry of the Earth: Practical 8
Stable Isotopes



Applications of Stable Isotopes

Practical 8:
Stable Isotopes

- Understand the terminology of stable isotope studies
- Investigate how stable isotope geothermometers are used to interpret formation of rocks and mineral deposits

STABLE ISOTOPES – BASIC TERMINOLOGY

Phase A ⟺ Phase B

$$\alpha_{A\text{-}B} = R_A / R_B$$

$$\alpha_{qz\text{-}mt} = (^{18}O/^{16}O)_{quartz} / (^{18}O/^{16}O)_{magnetite}$$

R is the ratio of the heavy to light isotope

$$\delta = (R_{sample} - R_{standard}) / R_{standard} * 1000 \quad ‰$$

$$\delta^{18}O = ((^{18}O/^{16}O)_{sample} - (^{18}O/^{16}O)_{SMOW}) / (^{18}O/^{16}O)_{SMOW} * 1000 \quad ‰$$

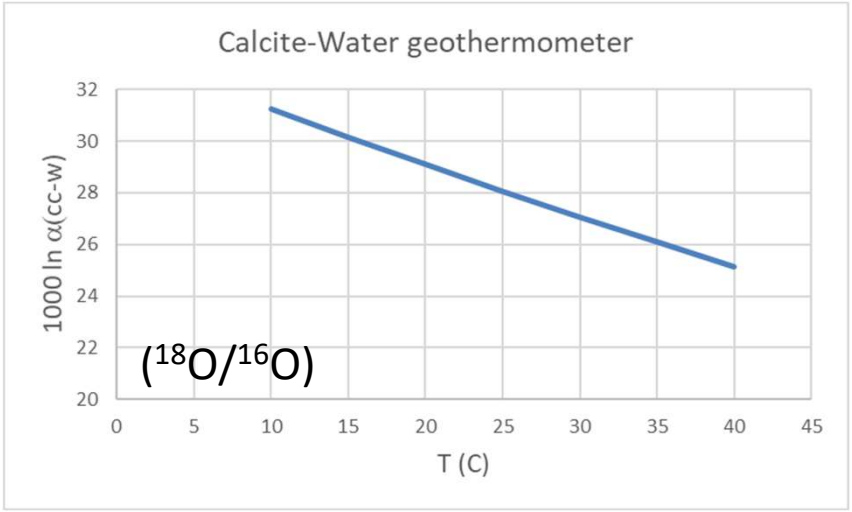
Putting δ together with α:

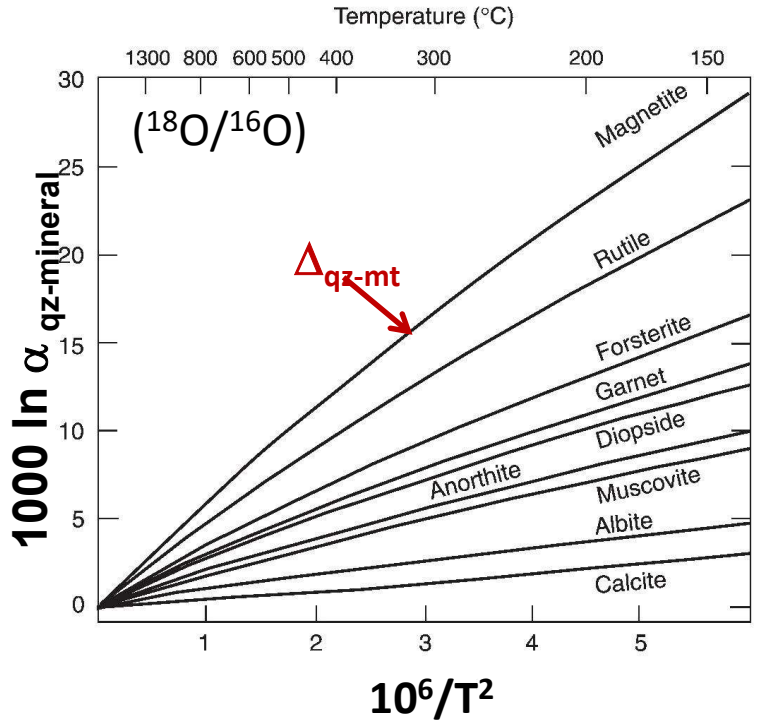
$10^3 \ln\alpha$ is approximately the same as $\delta_A - \delta_B$

$$10^3 \ln\alpha \approx \delta_A - \delta_B = \Delta_{A\text{-}B}$$

GEOTHERMOMETRY: Measuring the temperature of equilibria









https://www.sandatlas.org/magnetite/

Magnetite
& Quartz